Ads Notes

Y R

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1 Ranking Score

In ads ranking, the ranking score is often estimated as:

$$\text{Ranking Score} = \text{Bid} \times pCTR \times pCVR$$

which represents the expected value (e.g., expected revenue) per impression.

Auction: rank ads by score and determine which ads are shown and sometimes their prices. (展示广告的排序和位置)

User experiences will affect when considering pCTR.

(User experiences 会影响CTR, 但另一方面来说，不能说pCTR直接受user experience影响，因为“图片吸引人”或者“标题具有引导性”都会提成pCTR但是用户体验是negative的)

2 Commonly used feature engineering

- user features

  historical CTR, demographics, user embedding

- ad features

  creative(image, text, video features), ad history, ad category , historical performance

- Historical statistical features

  user CTR over 1/7/30 days.

  ad CTR by category , user-specific ad CTR

- cross features

  user clicked the ads before? (user $\times$ ad /category /advitiser)

  user-ad interaction frequency

- context features

    time, device, position, hour of the day, connection type

- features crossing for nonlinear relationship

    user age × ad category

3 常用训练模型

Increasing capability to model feature interaction

- Logistic Regression: Linear model, suitable for sparse features. Relay heavily on manual feature engineering
- LR+ feature crossing
- GBDT (Gradient Boosted Decision Trees): 优势： automatic fetaure partitioning, nonlinear interactions, no feature scaling.
  常用： XGboost, LightBGM
- GBDT + LR
- Deep CTR models: DeepFM / DCN (deep cross network). for low and high order interactions
- DIN (deep Interest Network): Models users interests from **historical behaviour sequences** using attention.
  eg: 点击历史，浏览历史，购买历史。
- DIEN/Transformer-based models

4 Cold Start

Modern approach: Content-based embeddings + representation learning

For new ads:

- content based features
  - categorical features: advertiser-ID, category , campaign type
  - creative features: image(handle with CNN or Version Transformer), text embedding (ad title or description)
  - format features: video/ image, color, placement(广告位),...
- leverage similar ads

  - build ad embedding share the same or similar category /advertiser-ID
  - use KNN in embedding space
- exploration strategy
  - user Thompson Sampling or Upper Confidence Bound (UCB) to initially show the ad to small and representative sample
  - start with prior CTR (like category average value) and update
  - For first 1000 impression, for example, on "exploration", then on real data.

For new user

- contextual signals and request time
  - Device models, OS, network type. time of day, location, languages, traffic source(from website/app/events...)
- lookalike model
  - nearest cluster of existing user
- rapid adaptation: 新用户虽然一开始没有历史数据，但只要产生少量行为，就应该尽快利用这些行为更新模型对该用户的理解，而不是一直依赖默认画像。
  - Continuously update user embeddings as interactions arrive.

Embeddings can be learned from historical click data using two-tower models, Word2Vec-style objectives, or deep CTR models.